RV College of Engineering

(An Autonomous Institution Affiliated to VTU)

R V Vidyanikethan Post
Mysuru Road Bengaluru - 560 059

III Semester B.E. Regular / Supplementary Examination February/March 2026

Bio Safety Standards and Ethics

Time : 3 Hours **Maximum Marks : 100**

Instructions to the students

1. Answer all questions from Part A . Part A questions should be answered in first three pages of the answer book only.
2. Answer Five full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

Part A

Question No	Question	M	CO	BT
1.1	List the roles of the legal and regulatory framework for biosafety levels in India.	02	2	1
1.2	Define GAP and how it helps in food production?	02	3	1
1.3	Give the role of different types of microorganisms in food spoilage	02	3	2
1.4	Give some examples of chemical presevatives and uses	02	4	1
1.5	Define Plagiarism and mention its implications	02	4	2
1.6	What is the primary purpose of a Biosafety Cabinet (BSC)?	02	2	1
1.7	List the roles of the legal and regulatory framework for biosafety levels in India.	02	2	1
1.8	State two general principles of food microbiology.	02	3	2
1.9	Give examples and advantages of smart or intelligent packing of food.	02	4	3
1.10	Give reasons for labeling of food additives and allergens.	02	3	2

Part B

Question No	Question	M	CO	BT
2a	Explain the construction, classification, working, and applications of Biosafety Cabinets in biotechnology and biomedical research.	08	2	3

2b	Explain the major differences in terms of air recirculation between various types of Class II Biosafety Cabinets.Draw a neat diagrm of class 2 biosafety cabinets.	08	2	4
3a	Farmers in the Northern part of India are protesting against the genetically modified Mustard seeds. In this context discuss in detail the pros and cons of the GM Mustard seeds.	16	2	3
	OR			
4a	Discuss in detail the Biosafety Guidelines of the Government of India and their alignment with international standards.	08	2	3
4b	Evaluate the role of Institutional Biosafety Committees (IBCs) and the Review Committee on Genetic Manipulation (RCGM) as per Indian biosafety guidelines.	08	2	5
5a	Elaborate on methods of food analysis add a note on the instruments used for analysis	08	3	3
5b	Elaborate on the sources and types of microbial contamination in food and the preventive strategies employed to control them.	08	3	5
	OR			
6a	Why food testing is considered important ? discuss the parameters and who will be the beneficiary of food testing.	08	3	4
6b	Give an account on industrial importance of beneficial microorganisms in food processing, with examples of organisms and end products.	08	3	3
7a	Discuss the different types of active and intelligent packaging methods and their applications.	08	3	4
7b	Elaborate on smart packaging and its importance in detail.	08	3	5
	OR			
8a	What are Good Production Practices (GPP) in the food industry? Discuss their importance.	08	3	3
8b	What are novel packaging materials, and how do they differ from traditional packaging materials?	08	3	3
9a	Elaborate on case study of Food Allergens, examples,symptoms,and treatment.	08	4	3
9b	What is plagiarism and which all activities were considered as plagiarism?	08	4	3
	OR			
10a	Elaborate on the the role of food preservation in food safety for the benefit of consumers	08	4	5
10b	Mention the factors which contributes to food borne illness? Elaborate on each of the factors.	08	4	4